

Let's use Black-Scholes on Tim's shares to calculate the default risk of the Opa:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

Where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Let's use Newton-Raphson to solve the equation, define $f(\sigma)$ as:

$$f(\sigma) = C_{BS}(\sigma) - C_{market}$$

We want σ^* such that $f(\sigma^*) = 0$.

So:

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{market}}{\text{Vega}(\sigma_n)}$$

Where:

$$\text{Vega} = S_0 \cdot N'(d_1) \cdot \sqrt{T}$$